SHI FENG

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EDUCATION

Tsinghua University *Sep 2019 - Jul 2023(expected)*

Bachelor of Science, selected to Yao Class

• GPA: 3.87/4.0

• GRE Scores: Verbal Reasoning 157, Quantitative Reasoning 170, Analytical Writing 4.0

• TOEFL Best Scores: Reading 30, Listening 29, Speaking 22, Writing 29

• Selected Courses: Introduction to AI (A), Mathematics for Computer Science (A), Causal and Statistical Learning (A+), Abstract Algebra (A), Introduction to Databases (A+), Artificial Intelligence: Principles and Techniques (A), Theory of Computation (A), Distributed and Blockchain System (A-), Machine Learning (A), Operating System (A), Research Immersion Training (A+)

SELECTED HONORS & AWARDS

"Star of Tomorrow" Award of Excellence *Mar 2022*, MSR Asia

Best Paper Award*(1/57)* *Nov 2021*, CSoNet

Innovation Excellence Award*(top 10%)* *Oct 2021*, THU

Social Practice Excellence Award *Oct 2021*, THU

"Challenge Cup" Science and Technology Competition, Top Grade Prize*(top 3%)* *Apr 2021*, THU

Comprehensive Excellence Award*(top 10%)* *Oct 2020*, THU

Russian Mathematical Olympiad, Silver Medal *Apr 2019*, MSE of Russia

Chinese Mathematical Olympiad, Gold Medal *Nov 2018*, CMS

National High School Mathematics League, First Prize *Sep 2016, 2017, 2018*, CMS

PUBLICATION

Shi Feng, Fang-Yi Yu, Yiling Chen. Peer Prediction for Learning Agents. **NeurIPS** 2022. **[arXiv] [code]**

Shi Feng, Wei Chen. Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model. **International Conference on Computational Data and Social Networks (CSoNet)** 2021. **[arXiv] [DOI]**

Shi Feng, Wei Chen. **Combinatorial Causal Bandits**. In submission. **[arXiv] [code]**

RESEARCH INTERESTS

Theoretical computer science and economics, especially in algorithmic game theory, learning theory, causality, and machine learning.

RESEARCH EXPERIENCES

Multi-Agent Owner-Assisted Scoring Mechanisms *Aug 2022 - now*

Advisors: Prof. **Yuhao Wang**, Prof. **Weijie Su** Tsinghua IIIS&Upenn Wharton School

• Extended owner-assisted scoring problem to multi-agent setting and wrote a research proposal for it.

• Designed a fair, "exchangeable noise assumption"-free, and truthful mechanism for the extended problem.

Peer Prediction for Learning Agents *Feb 2022 - Jun 2022*

Advisors: Prof. **Yiling Chen**, Prof. **Fang-Yi Yu** Harvard EconCS Group

• Surveyed mechanisms for information elicitation and extended the original multi-task setting to a sequential setting.

• Proved truthful convergence of correlated agreement mechanism adopted on agents using reward-based online learning algorithms and a negative result showing no-regret assumption on agents can not guarantee truthful convergence.

• Created a peer prediction simulator and conducted numerical experiments which supported theoretical results.

Combinatorial Causal Bandits *Jul 2021 - Feb 2022*

Advisor: Prof. **Wei Chen** MSR Asia Theory Group

• Surveyed causal bandits algorithms for pure exploration and cumulative regret minimization.

• Proposed the new combinatorial causal bandit framework with a definition of cumulative regret.

• Proposed an algorithm for combinatorial action space bandits problem on Markovian binary generalized linear models.

• Proposed an algorithm for combinatorial action space bandits problem on binary linear models with unobserved nodes.

• Implemented algorithms we proposed in OOP and conducted numerical experiments to show their efficiency.

Causal Inference for Social Networks *Nov 2020 - Jul 2021*

Advisor: Prof. Wei Chen MSR Asia Theory Group

• Surveyed identifiability of Bayesian causal models and completeness of do-calculus algorithm.

• Provided transformations from an Independent Cascade Model to a Bayesian causal model.

• Proved necessary and sufficient conditions for identifiability in three common types of Independent Cascade Models.

TALKS

NeurIPS 2022, Peer Prediction for Learning Agents *Oct 2022*
Yao Class Seminar, Peer Prediction for Learning Agents *Sept 2022*
CSoNet 2021, Causal Inference for Influence Propagation–Identifiability of the Independent Cascade Model *Nov 2021*

SERVICES

Class Leader, Yao Class 91, Tsinghua University *Aug 2022 - now*
Reviewer, Journal of Combinatorial Optimization (JOCO) *May 2022*
Team Member, Chinese national team for Russian Mathematical Olympiad *Apr 2019*

SKILLS

Programming Languages: C/C++, Python, Golang, SQL, Assembly language, Bash
Professional Applications: Latex, Mathematica, Matlab, PyTorch, TensorFlow, Git
Languages: Chinese (native language), English (fluent)
Hobbies: soccer, road trip, handball, basketball, driving